

HUMAN-SUPERVISED DISASTER ROBOTICS UNDER UNRELIABLE COMMUNICATION

A System-Level Engineering Framework for Resilient Disaster Response

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THE PROBLEM

During disasters, communication networks, GPS signals, and environmental perception degrade or fail simultaneously. Existing robotic systems—whether teleoperated or autonomous—struggle under these coupled failures, leading to loss of situational awareness, poor decision making, and mission failure when robots are needed most.



OUR CONTRIBUTION

This research proposes a **system-level engineering framework** for human-supervised semi-autonomous disaster robots that maintains operational continuity under unreliable communication.

The framework reframes the problem from optimizing individual components to engineering for **graceful degradation under coupled system failures**.

SYSTEM-LEVEL ENGINEERING FRAMEWORK



Mission failure occurs when any or multiple subsystems (communication, localization, perception, planning, human awareness) degrade beyond critical levels. Our framework ensures the system **degrades gracefully** and continues to execute the mission.



EXPERIMENTAL EVALUATION

The proposed framework was evaluated through a simulated disaster-response system (ROS-inspired modular architecture) incorporating:

- SLAM-based localization and mapping
- Communication-aware shared-autonomy control
- Persistent environmental memory
- Simulated degraded communication conditions



Key Results: The system demonstrated mission continuity, persistent mapping, reduced dependence on continuous teleoperation, and graceful degradation instead of abrupt system failure under degraded communication.



REQUEST: I would be honoured for an opportunity to briefly present this research to your organisation's technical or advisory team, receive expert feedback, and better understand the operational challenges faced during real-world disaster response.



KEY TAKEAWAYS



Communication-Aware Shared Autonomy

Enables robots to continue safe operation despite intermittent connectivity, high latency, and communication loss, while allowing human supervisors to intervene when required.



Resilient Localization and Mapping

Combines **SLAM**, multi-sensor fusion, and onboard environmental memory to operate reliably in GPS-denied and infrastructure-damaged environments.



Failure-Resilient System Design

A coupled-systems perspective that prioritizes graceful degradation and operational continuity rather than peak performance under ideal conditions.



OPERATIONAL RELEVANCE

The framework may inform future engineering approaches for:

- Urban Search and Rescue (USAR)
- Earthquake and structural-collapse response
- Flood and landslide operations
- Industrial accident response
- Hazardous or GPS-denied environments
- Human-robot collaboration during emergencies



Full Research Paper:
Human-Supervised Disaster Robotics Under Unreliable Communication
(Available upon request)